

Question	Answer
What SLURM is and its role in HPC clusters.	Slurm (Simple Linux Utility for Resource Management) is a open source manger for compute resources acting as a scheduler for tasks based on the needs of said tasks, their priority and existing resources.
Login Nodes vs Compute nodes	Login nodes are for user operated tasks such as editing code, file management and job submission. Compute nodes are for simulations and other heavy duty usages, including ai, once submitted to the job queue.
Job Scripts and SBATCH	A job script is the method used to allocate resources to a program, to associate who the program is from and what it should do during the allocated space to operate. SBATCH is the command type to submit a non interactive job script to the slurp manager.
Requesting resources	Resources are requested as per the job script in sbatch, asking for memory, notes and cpu and gpu usage, as seen below. <pre>#SBATCH --mem=1G # memory required per node, in the form of [num][M G T] #SBATCH --ntasks=1 # number of Slurm tasks to be launched, increase for multi-process runs ex. MPI #SBATCH --cpus-per-task=2 # number of processor cores to be assigned for each task, default is 1, increase for multi-threaded runs #SBATCH --ntasks-per-node=1 # number of tasks to be launched on each allocated node # create job array of 10 tasks numbered 1 through 10</pre>
Job types	Serial-Uses a single cpu core which is best for simple or non parallel jobs(must be completed in sequence) Array-Submission of multiple variations of a serial job for a parameter sweep Multinode-For pallelized jobs that need to run simultaneously for large scale taks
Submitting jobs	By using sbatch, a job can be submitted based on its job script to properly allocate itself in the queue as needed. It will attempt to run based on its priority and allocation needs.
Monitoring jobs	Jobs can be monitored by using Squeue which shows the current jobs waiting(as PD) or running(as R), their time of compute, the nodes they require or why they are waiting(priority or resources). Sinfo gives general information on the nodes themselves showng they are being allocated or not and what recourses nodes have in total.
Job states and managing	States detail how the job has operated and if has completed or failed. PD-Pending, R-Running, CG-Completed successfully S-Suspended, F-Failed, OOM-Out of memory, CA-Cancelled by user. Squeue allows you see the current queue of jobs, scontrol allows you to modify and see advance details of any job, scancel can kill any job for any reason.

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Job arrays and dependencies	A job array is able to run similar or identical tasks in parallel to reduce overall compute time, meanwhile a dependency makes jobs operate in the correct sequence, so jobs can rely on other jobs being completed.
SLURM environment variables	Environment variables provide information to the compute job as it runs and are assigned automatically. This allows for closer monitoring of the given jobs
Job accounting with sacct	Sacct allows to see the accounting and resource data of the jobs in slurm, which allows you to see their current state. It is best used for history monitoring and efficiency monitoring of completed jobs as it acts as a database for said jobs.
Best practices and pitfalls	All long jobs must be run on compute nodes as login nodes are often shared. Small jobs should be stacked into bigger jobs to reduce overhead Avoid creating too many small files. -exclusive takes up a full node, so only use if you need all resources on a node -exclusive also causes your program to wait longer Checkpoints allow your code to recover if possible. Using them means the job can run again separately Predicting how long the job will run before submission makes running it less likely to time out.